Hiten Dalmia

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Education

Chennai Mathematical Institute — Chennai, India
Master of Science in Theoretical Computer Science (On Full Scholarship) — *Aug. 2026 – Discontinued*

Kirori Mal College, University of Delhi — Delhi, India
Bachelor of Science (Honours with Research) in Mathematics, Minor in Economics — *Nov. 2022 – May 2026*

Experience

Python Developer — Mar. 2023 – Present
Freelance & Fractal — *Remote*

- Delivered 10+ production applications for client businesses via Upwork and Fractal, specializing in Python backend architecture, RESTful APIs, document processing pipelines, and relational database design.
- **Order Procurement & Document Processing Solution** | *Python, PostgreSQL, RESTful APIs, PIL, AWS*
  - Designed a modular Python order and document management backend using PostgreSQL, session authentication, and RESTful APIs to process client transactions.
  - Constructed a dual-tier administrative platform with role-based access, real-time inventory tracking, employee task dispatch, audit logging, and automated recurring order schedules.
  - Built an automated Python document extraction pipeline using Optical Character Recognition (OCR) and PIL to parse incoming WhatsApp order documents into structured PostgreSQL records.
  - Optimized relational PostgreSQL database schemas with B-tree indexing and foreign key integrity constraints, sustaining peak query loads under 50 ms.

Computer Vision Engineer — Oct. 2023 – Apr. 2025
Tagglabs Experimental Pvt. Ltd. (Contract) — *Gurugram, Haryana (Remote)*

- **Cross-Platform Computer Vision AR App** | *Python, OpenCV, PyTorch, C++, Django* **(Live link)**
  - Integrated OpenCV Scale-Invariant Feature Transform (SIFT) keypoint extraction into a cross-platform C/C++ module, unifying mobile computer vision execution across Android and iOS.
  - Developed a real-time image processing pipeline using PIL and OpenCV to normalize OS camera frame buffers and aspect ratios into raw C structs, maintaining zero memory leaks at 30+ FPS.
  - Deployed PyTorch deep learning models optimized via ONNX Runtime and FP16 quantization for real-time keypoint matching, face detection, and object recognition.
  - Built RESTful backend APIs in Django hosted on AWS Lambda to process spatial discovery requests, image upload payloads, and feature vector lookups for alpha release.
- **Dealer Event Platform** | *Python, Django, Flutter, RESTful APIs*
  - Created an event platform for 500+ attendees at the Kia National Conference in Bangkok, pairing a Flutter frontend with a Python Django backend on AWS Lambda for real-time attendee registration.

Technical Skills

Languages & Core Systems: Python, C++, C, SQL, TypeScript/JavaScript, Dart, R, Mathematica
AI, ML & Image Processing: OpenCV, PIL, PyTorch, TensorFlow, ONNX, OCR, Quantisation, NumPy, Pandas
Frameworks & Cloud: Django, RESTful APIs, PostgreSQL, AWS (Lambda, Serverless), Docker, Flutter, NextJS
Developer Tools & Methods: Git, Linux, Bash, Vim, LaTeX, CI/CD, Agentic Development, Context Engineering

Achievements

Erasmus Mundus Joint Master MATHS-DISC Fellowship — Europe
Issued by European Union and affiliated universities — *Mar. 2026*

Master in Mathematics (ISI) and MSc Computer Science (CMI) Scholarships — India
Indian Statistical Institute and Chennai Mathematical Institute respectively — *Jun. 2026*

Publications

Computable Analysis: Mathematical Foundations and Computability of Real Functions
Poincare Journal of Mathematical Physics & Applications, Vol. 1, No. 1 (2026), 1–20 (with Sujit Kumar)